SIGNIFICANCE TEST FOR THE TALBURT-WANG SIMILARITY INDEX

(Research Paper)

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The Talburt-Wang Index (TWI) has proved to be a useful tool for assessing the outcomes of customer data integration (CDI) processes. However in practices, the question often arises as to whether the TWI value indicates any "significant difference" between two CDI outcomes. This paper proposes a method for calculating a measure of the significance of the variation indicated by the TWI values. The method (1) identifies the extreme values (the best and the worst values) for a quality index called Q, (2) maps the TWI on this range, and (3) calculates level of significance for TWI.

Keywords: Customer Data Integration, Data Quality, Quality Index, and Significance Measure

INTRODUCTION

Let us assume that there are N records representing K customers (N >> K). The process that maps the N records into the K customers is referred to as the *customer data integration (CDI)* process. In our previous experiment [1], (1) we obtained a large number of records from mining of the "obituary" announcements in the on-line newspapers of the United States and (2) tried to integrate the obtained records with a customer database for a given enterprise [2]. The methodology for the mapping was a hybrid one using both "weight-based", and "fuzzy-based" approaches.

The main concern about the customer data integration, regardless of the methodology, is the quality of the outcome. One can assume that the integration (mapping) process may be conducted by one or more than one existing methodologies. Each methodology prescribes a set of heuristic or otherwise rules for the mapping. As a result, the mapping process may produce different outcomes using different methodologies and each outcome is simply a grouping of the N records such that each group potentially maps on one customer record.

Several tools have been suggested by [3, 5, 6, 10] and others for measuring the quality of integrations. The Talburt-Wang Index (TWI) presented in [8] is the simplest and most computationally effective among these tools in keeping with the notion of “data quality algebra” as introduced by [9]. This fact is more pronounced when the number of records is too large. One may ask the following intriguing question about TWI: When do you interpret the metric value as being significant? In this paper, we try to provide a formal approach to answer this question.

BACKGROUND

Given a set of customer records and two methods for grouping the records into groups belonging to the same customer, the application of each grouping method will result in a partition of the record set. The TWI will produce a value in the interval (0, 1] that indicates the degree of similarity between the partitions. The value of the TWI is 1 if and only if the two groupings being compared are identical. The less similar the partitions, the closer the value of the TWI moves toward zero.

In the case where the TWI value is 1 or the value is small, the interpretation of the value is relatively straightforward, i.e. the outcomes are identical or very different, respectively. However, for values close to 1, the issue is less clear. In a given context, what is the threshold value of the TWI that indicates that two outcomes were significantly different – smaller than 0.99, 0.95, 0.9, or even smaller? This is often an important question when deciding whether a change in a CDI process is providing any significant improvement, especially if the change is costly in terms of memory or complexity.

THE INTERSECTION MATRIX

In a CDI process, let A and B represent the two groupings of the records generating the partitions $\{A_1, . . ., A_m\}$ and $\{B_1, . . ., B_n\}$, respectively. Thus, the intersection matrix for A and B groupings has m rows and n columns. Furthermore, let r_i be the number of non-empty cells in the i-th row, and let c_j be the number of non-empty cells in the j-th column of the intersection matrix. Table 1 shows a typical intersection matrix where A has 4 groupings and B has 6 groupings.

A/B	B1	B2	B3	B4	B5	B6	r
A1	3		2			1	3
A2		5					1
A3		1		2	1		3
A4			4			3	2
c	1	2	2	1	1	2	**9**

Table 1: Typical Intersection Matrix

The numbers in the cells show the size of the intersection between two groupings. For example, A3 and B4 have a common intersection of two elements. If there is no intersection, the cell is empty. The values in the row labeled with "c" are the counts of non-empty cells in the column (c_j), and the values in the column labeled "r" are the counts of the non-empty cells in the row (r_i). The value in the lower right-hand corners (9) represents the total number of overlaps between the two partitions.

THE TWI INDEX

The calculation of the TWI is relatively straightforward. It is the product of the number of groups in two partitions divided by the square of the number of overlaps between the two partitions.

$R = \Sigma(r_i,)$ for i = 1 to m
$C = \Sigma(c_j,)$ for j = 1 to n

Note that

$R = C$

And is the total number of overlaps between A and B.

$TWI = (|A| * |B|) / R^2$

THE Q INDEX

In order to determine the significance for TWI, we first introduce a new index (called Q), and then establish the relationship between Q and TWI.

Let us define Q as follows:

$Q = (|A| + |B|) / (R + C)$

or

$Q = (m + n) / 2R$ **Formula I**.

The case in which A and B groupings are exactly the same is considered to be the best case because Q = 1. (|A| = |B|, R = |A| and thus Q = 1.)

A/B	B1	B2	B3	B4	B5	B6	r
A1	2						1
A2		2					1
A3			1				1
A4				2			1
A5					1		1
A6						1	1
c	1	1	1	1	1	1	**6**

Table 2: An intersection matrix for the best case.

Applying Formula I, the best value for Q is

$Q_{best} = (m + m) / 2R = m/R = |A|/R$ **Formula II.**

The intersection matrix for A and B groupings shown in Table 2 is an example of the best case scenario in which R = C = m = 6, and $Q_{best} = m/R = 6/6 = 1$.

The similarity between A and B groupings is the worst when A grouping has m groups (m > 1) while B has only one group. In this case, |A| = m and |B| =1. Using Formula I,

$Q_{worst} = (|A| + 1)/2R = (m + 1)/2R$ **Formula III**

The intersection matrix for A and B groupings shown in Table 3 is an example of the worst case scenario in which R = 6, m = 6 and $Q_{worst} = (6 + 1) / (2*6) = 7/12$.

A/B	B1	r
A1	2	1
A2	2	1
A3	1	1
A4	2	1
A5	1	1
A6	1	1
c	6	**6**

Table 3: An intersection matrix for the worst case.

For a given intersection matrix that includes A and B groupings, one may use Formulas I, II and II to get the index (Q), the upper limit (Q_{best}) and the lower limit (Q_{worst}) of this index and then compare Q with the two limits and determine whether the Q is significant. If Q is the outside of range [Qbest – Qworst] , then Q is not significant; Otherwise, Q is significant and the significance level of Q is calculated using the following formula:

$S = (Q – Q_{worst}) / (Q_{best} – Q_{worst})$, where $0 \leq S \leq 1$. **Formula IV**

When S moves closer to one, then significance level increases.

Now consider a more typical intersection matrix as shown in Table 4 which is neither the best nor the worst case scenario:

R = 9; m = 9; n = 7

$Q_{best} = |A|/R = 9/9 =1$

$Q_{worst} = (|A| + 1) / 2R = (10/18) = 0.56$

Q = (|A| + |B|) / 2R = (9 + 7) / (2*9) = (16) / (18) = 8/9 = 0.89

S = (Q-Qworst) / (Qbest – Qworst)

= (0.89 – 0.56) / (1-.56) = 0.33/0.44 = 0.75

A/B	B1	B2	B3	B4	B5	B6	B7	r
A1	1							1
A2		1						1
A3			2					1
A4	1							1
A5				1				1
A6			1					1
A7					1			1
A8						1		1
A9							1	1
c	2	1	2	1	1	1	1	**9**

Table 4: Typical Intersection Matrix

Let us consider the transposition of the intersection matrix of Table 4 for which m=7, n=9 and R = 9. The Q_{best} = 7/9 = 0.78 and Q_{worst} = 8/18 =0.44, and Q = 16/18 = 0.89. In this case the quality index Q is not significant because it is outside of the range [Q_{best} - Q_{worst}]. Obviously there is a contradiction between the two interpretations of Q for Table 4. To solve this problem we get help from an information theoretic approach.

Let us concentrate on grouping A. One may consider grouping A as classification of a set of records. In Information Theory [4, 7], the expected information content of the grouping A of the customer records is given by the following formula:

$$E(A) = \Sigma(-P^i \log_2^{Pi}) \qquad \text{for } (i = 1 \text{ to } m) \qquad \textbf{Formula V}$$

Where P^i is the probability for group A_i and it is approximated by the relative frequency of the records in A_i. If the total customer records is N, then $P^i = |A_i|/N$.

To solve the contradiction problem for Table 4, we calculate both E(A) and E(B) and the one with the higher expected information is the winner and its cardinality is used for value of m in calculation of Q_{best} and Q_{worst}.

Now let us go back to the example of Table 4 and decide what the value of m is. The A grouping generates nine groups of A1, A2, A3, A4, A5, A6, A7, A8, and A9 out of the 10 customer records with probabilities of 0.1, 0.1, 0.2, 0.1, 0.1, 0.1, 0.1, 0.1, and 0.1, respectively.

$$E(A) = -0.1 \log_2^{0.1} - 0.1 \log_2^{0.1} - 0.2 \log_2^{0.2} - 0.1 \log_2^{0.1} - 0.1 \log_2^{0.1} -0.1 \log_2^{0.1} - 0.1 \log_2^{0.1} - 0.1 \log_2^{0.1} - 0.1 \log_2^{0.1}$$
$$= 0.332 + 0.332 +0.464 +0.332 +0.332 +0.332 +0.332 +0.332 +0.332$$
$$= 3.12$$

The B grouping generates seven groups of B1, B2, B3, B4, B5, B6, and B7 out of the 10 customer records with probabilities of 0.2, 0.1, 0.3, 0.1, 0.1, 0.1, and 0.1, respectively.

$$E(B) = -0.2 \log_2^{0.2} - 0.1 \log_2^{0.1} - 0.3 \log_2^{0.3} - 0.1 \log_2^{0.1} - 0.1 \log_2^{0.1} -0.1 \log_2^{0.1} - 0.1 \log_2^{0.1}$$
$$= 0.464 + 0.332 + 0.521 + 0.332+ 0.332+ 0.332+ 0.332$$
$$= 2.645$$

Since E(A) > E(B), then A grouping is the winner and m = 9.

WHAT Q AND SIGNIFICANCE CALCULATION HAS TO DO WITH TWI?

We try to establish a relationship between Q and TWI (for the sake of simplicity, we refer to TWI as T) in this section. As we have expressed before,

$Q = (m + n) / 2R$ and
$T = (mn)/R^2$
$Q = TR(m+n) / 2mn$ **Formula VI**

When T is given, then formula VI can be used to convert it to its equivalent Q (we call it Q_t) and then Q_t will be compared to Q_{best} and Q_{worst} to determine the significance of T.

The value of Q_t may be located inside or outside of the range [$Q_{best} – Q_{worst}$]. If it is outside of this range, then T is not significant; otherwise, it is significant. The significance level of T may be calculated using formula IV.

Example: Let us use the example in Figure 1 as given in the paper [8]. There are 25 records in this example.

The A and B groupings create 4 and 6 partitions, respectively. The Expected information for A and B groupings are:

$E(A) = -6/25 \log_2^{6/25} – 10/25 \log_2^{10/25} – 3/25 \log_2^{3/25} – 6/25 \log_2^{6/25} = 2.118$

$E(B) = -2/25 \log_2^{2/25} – 4/25 \log_2^{4/25} – 10/25 \log_2^{10/25} – 2/25 \log_2^{2/25} – 4/25 \log_2^{4/25}$
$– 3/25 \log_2^{3/25} = 7.563$

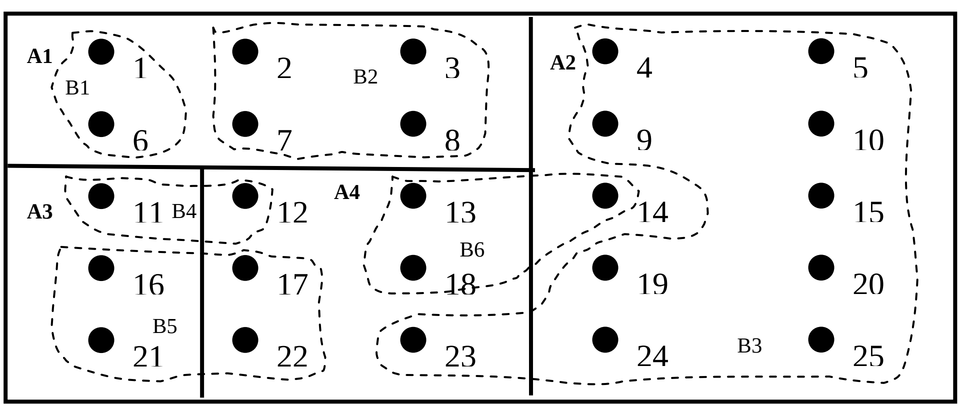

Figure 1: A customer file with 25 records and two groupings of A and B.

Therefore, the winner is B grouping and thus m = 6. The intersection matrix is shown in Table 5.

B/A	A1	A2	A3	A4	r
B1	2				1
B2	4				1
B3		9		1	2
B4			1	1	2
B5			2	2	2
B6		1		2	2
c	2	2	2	4	**10**

Table 5: The intersection matrix for Figure 1.

$m = 6$; $n = 4$; $R = 10$;
$Q_{best} = m/R = 6/10 = 0.6$
$Q_{worst} = (m+1)/2R = 7/20 = 0.35$
$T = (mn)/R^2 = 24/100 = 0.24$
$Q_t = TR(m+n) / 2mn = (0.24)(10)(4 + 6) / 2*4*6 = 0.5$;

Consequently, we conclude that T is significant because it is in the range of [0.6 – 0.35] and its significance level is:

$S = (Q_t – Q_{worst}) / (Q_{best} – Q_{worst})$
$= (0.5 – 0.35) / (0.6 –.35) = 0.15/0.25 = 0.6$

SUMMARY AND FUTURE RESEARCH

When the TWI(A,B) = 0.78, for example, does it mean that the A and B groupings are significantly different or more-or-less equivalent? The intriguing question is: "*When do you interpret the TWI value as significant?*" An answer to this question was provided in two steps: (1) Formally defining significance range of values [Q_{best}, Q_{worst}] for a different quality index (called Q) using the CDI groupings of A and B and (2) mapping the TWI on this range and calculate its level of significance.

As future research, we are working on estimation of accuracy for a given CDI grouping. The similarity significance may play a role in this new research effort.

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